

Steering Guidance for Personalized Text-to-Image Diffusion Models

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Figure 1. Comparison of generated image quality between Classifier-Free Guidance (CFG) and our Personalization Guidance.

Abstract

Personalizing text-to-image diffusion models is crucial for adapting the pre-trained models to specific target concepts, enabling diverse image generation. However, fine-tuning with few images introduces an inherent trade-off between aligning with the target distribution (e.g., subject fidelity)

and preserving the broad knowledge of the original model (e.g., text editability). Existing sampling guidance methods, such as classifier-free guidance (CFG) and autoguidance (AG), fail to effectively guide the output toward well-balanced space: CFG restricts the adaptation to the target distribution, while AG compromises text alignment. To address these limitations, we propose personalization guidance, a simple yet effective method leveraging an unlearned weak model conditioned on a null text prompt. Moreover, our method dynamically controls the extent of unlearning

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in a weak model through weight interpolation between pre-trained and fine-tuned models during inference. Unlike existing guidance methods, which depend solely on guidance scales, our method explicitly steers the outputs toward a balanced latent space without additional computational overhead. Experimental results demonstrate that our proposed guidance can improve text alignment and target distribution fidelity, integrating seamlessly with various fine-tuning strategies.

1. Introduction

Diffusion models [11, 32] have emerged as powerful generative models capable of representing complex data distributions, driving significant advancements in high-fidelity image generation [24, 26, 29, 38]. However, real-world use cases often demand personalized generation, where the model generates not only generic concepts but also specific concepts like a novel character or person [8, 20, 27]. This personalization technique is particularly useful for applications such as portrait editing and specific content creation.

Many approaches have been proposed to personalize diffusion models for novel concepts. One line of work [6, 28, 30, 34] learns a general-purpose encoder capable of adapting to new concepts without test-time finetuning. However, this approach requires large-scale text-image datasets and considerable computational resources, and must often be re-trained from scratch for each pre-trained model, making it expensive to adapt to the growing variety of text-to-image or text-to-video diffusion architectures. An alternative solution is to directly fine-tune a pre-trained diffusion model on the target images, which can be roughly categorized into fine-tuning low-rank adapters [5, 13], text embeddings [8], specific layers [14, 20], or the full model parameters [27]. This strategy can be applied to any pre-trained diffusion model and only requires a few target images, proving effective for personalization in various scenarios.

Despite these advancements, optimization-based personalization remains a nontrivial challenge. When fine-tuning a model to a specific, often much smaller, target data distribution, there is an inevitable tradeoff between aligning with the target distribution (*e.g.*, learning a novel concept with a few target images) and preserving broad knowledge of the original model (*e.g.*, preserving text editability). This trade-off manifests in various ways: a fine-tuned model for personalization may generate high-quality domain-specific images but lose text editability, making it difficult to generalize across different prompts. Conversely, a model that retains broad knowledge from its original training distribution may struggle to fully align with new concepts introduced in fine-tuning. Existing approaches attempt to mitigate this trade-off by adjusting the number of fine-tuning iterations [8, 27], leveraging regularization [14, 27], reducing the model pa-



Figure 2. Motivation of Personalization Guidance. Classifier-free guidance [10] can interfere with subject fidelity due to its alignment with the text prompt. In contrast, autoguidance [18] fails to reflect the text prompt but generates the subject more similarly. Our guidance complements the limitations of both methods.

rameters of updates [5, 13], yet they primarily operate at the fine-tuning stage. Moreover, balancing the trade-off between comprehensive text editability and precise alignment with a new target distribution remains an open problem.

A parallel line of research focuses on sampling guidance techniques, such as Classifier-Free Guidance (CFG) [10] and AutoGuidance (AG) [18]. These methods rely on “weak models” to detect and push the outputs away from poor trajectories, thereby guiding samples toward higher-quality regions of the data manifold. For instance, CFG uses an unconditional model using the null text prompt as the weak model, while AG uses an unlearned model—one with fewer parameters or less training—conditioned on the same text prompt. However, these methods have limitations when applied to personalized diffusion models, as shown in Fig. 2. CFG can interfere with the model’s adaptation to the target distribution, restricting its ability to fully utilize fine-tuned knowledge. Moreover, while the guidance scale of CFG adjusts the degree of text fidelity, it does not improve alignment with the target data. On the other hand, AG, which primarily enhances alignment with the target distribution, however, lacks an explicit mechanism for improving text alignment, making it unsuitable for fine-tuned text-to-image models.

To address these issues, we propose a simple yet effective guidance technique, called **Personalization Guidance**, for personalized diffusion models, which can dynamically balance text alignment and adaptation to the target distribution. Our method leverages an “unlearned” weak model with a null text prompt, without incurring additional computational overhead during inference. Here, we aim to balance the alignment with the target distribution (*e.g.*, subject fidelity) and the preservation of the broad knowledge of the original model (*e.g.*, text fidelity) by controlling the extent of “unlearning” in a weak model. Here, the degree of unlearning refers to the extent to which the target data distribution has not been learned compared to the fine-tuned model. The rationale behind adjusting this degree of unlearning is to steer the output toward an optimal, well-balanced latent space. Specifically, during inference, we interpolate the weight parameters between the pre-trained and fine-tuned

diffusion models to derive an optimal weak model, effectively controlling the extent of unlearning. Unlike guidance scales, which adjust the intensity of guidance, weight interpolation for the weak model serves to steer outputs in an appropriate direction. This straightforward approach can be readily integrated into various fine-tuning diffusion models.

2. Related Work

Personalization of Diffusion Models. Personalizing diffusion models aim to generate novel images by adapting to specific concepts using a set of reference images. Optimization-based methods personalize models by fine-tuning textual embeddings [8, 14, 20], entire model parameters with regularization [27], or low-rank adapters [5, 13]. Another line of research leverages extensive text-image datasets to train dedicated text-to-image personalization models, by fine-tuning additional modules [6, 28, 30, 34]. Recently, several studies [4, 22, 31] have achieved better personalization without additional fine-tuning or training. Orthogonal to these existing approaches, we introduce a simple yet effective personalization guidance technique that seamlessly integrates with various personalization methods by modifying the classifier-free guidance. Moreover, our guidance enables us to adjust the degree of subject and text fidelity without incurring additional computational costs.

Guidance for Diffusion Models. Classifier-Free Guidance (CFG) [10] enhances conditional image generation by leveraging the unconditional model as a weak model to guide the outputs toward better adherence to the given condition. Various guidance techniques have been explored, such as the guidance using perturbations [1, 16] or blurring to the attention maps [12]. Autoguidance [18] improves generation by using a weak model—one with fewer parameters or less training—to push outputs toward a well-trained direction. While InstructPix2Pix [2] introduces separate CFG using both image and text conditions, it cannot be directly applied to DB-LoRA due to its reliance on image-conditioned training. For personalization, subject-agnostic guidance [4] enhances subject-agnostic attributes by employing a weak model conditioned on subject-specific text prompts. In this study, we aim to address the fundamental personalization trade-off—aligning a model with a smaller target data distribution while preserving the broad text fidelity of the original model. We derive a weak model that dynamically adjusts the degree of adaptation, balancing comprehensive text editability with precise alignment to new concepts.

Weight Interpolation. There are previous studies on exploring the use of weight interpolation. WISE-FT [36] improves the robustness of fine-tuned classifiers against distribution shifts through weight-space ensembling. Model-soups [35] maximizes model accuracy and robustness by interpolating the weights of multiple fine-tuned classifiers trained with different hyperparameter configurations.

3. Method

3.1. Preliminaries

Diffusion Models. Diffusion models [11, 32] aim to learn reversing a forward noising process that gradually transforms data samples into noise. Concretely, the forward process starts with a data distribution $p_0(\mathbf{x})$ and ends at a high-noise distribution $p_T(\mathbf{x}) \approx \mathcal{N}(\mathbf{0}, \mathbf{I})$. Sampling from the data distribution proceeds by solving the equivalent probability-flow ordinary differential equation (PF-ODE). Under the variance-preserving parameterization where $p(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_0, \sigma_t^2\mathbf{I})$, the PF-ODE takes the form:

$$d\mathbf{x}_t = -\sigma_t \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) dt, \quad \mathbf{x}_T \sim p_T(\mathbf{x}_T). \quad (1)$$

In diffusion models, the score function is parameterized by ϵ_θ . Here, the noise estimation of the diffusion model can be considered as $\epsilon_\theta(\mathbf{x}_t) \approx -\sigma_t \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)$.

Personalization. Suppose we have a pre-trained diffusion model ϵ_θ , where θ is the parameters of the model. We wish to personalize it to a smaller target dataset $\mathcal{D}_{\text{target}} = \{(\mathbf{x}^i, \mathbf{c}^i)\}_{i=1}^N$, where each $\mathbf{x}^i$ and $\mathbf{c}^i$ is an image and its associated text prompt, respectively. The goal is to obtain new parameters θ' so that the fine-tuned model $\epsilon_{\theta'}$ can generate samples more faithfully matched the $\mathcal{D}_{\text{target}}$.

Fine-tuning typically follows the same denoising objective used by the original diffusion model [11]. Concretely, let σ be a noise level sampled from a training schedule, and let $\epsilon \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I})$. For each $(\mathbf{x}, \mathbf{c})$ in $\mathcal{D}_{\text{target}}$, we form the noisy input $\mathbf{x}_t = \mathbf{x} + \epsilon$ and minimize:

$$\mathcal{L} = \mathbb{E}_{(\mathbf{x}, \mathbf{c}) \sim \mathcal{D}_{\text{target}}, \sigma, \epsilon} \left\| \epsilon_{\theta'}(\mathbf{x}_t, \sigma, \mathbf{c}) - \epsilon \right\|_2^2. \quad (2)$$

This teaches the model to reconstruct $\mathbf{x}$ from noisy inputs in the domain of $\mathcal{D}_{\text{target}}$, thereby adapting it to the new concept. In practice, additional regularization terms are often used to prevent overfitting [14, 27].

Classifier-Free Guidance. To enhance the generation towards an arbitrary condition $\mathbf{c}$, classifier-free guidance (CFG) [10] introduces a new sampling distribution $p^\lambda(\mathbf{x}|\mathbf{c})$ composed with both $p(\mathbf{x})$ and the classifier distribution $p(\mathbf{c}|\mathbf{x})^\lambda$, which is described as:

$$p^\lambda(\mathbf{x}|\mathbf{c}) \propto p(\mathbf{x}) p(\mathbf{c}|\mathbf{x})^\lambda, \quad (3)$$

where $\lambda > 1$ amplifies the conditional likelihood. By applying Bayes' rule for some timestep t , the corresponding score function is:

$$\begin{aligned} \nabla_{\mathbf{x}_t} \log p^\lambda(\mathbf{x}_t|\mathbf{c}) &= \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t) \\ &+ \lambda (\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t|\mathbf{c}) - \nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t)). \end{aligned} \quad (4)$$

To enable conditional generation such as text prompts and class labels, CFG jointly trains the unconditional model

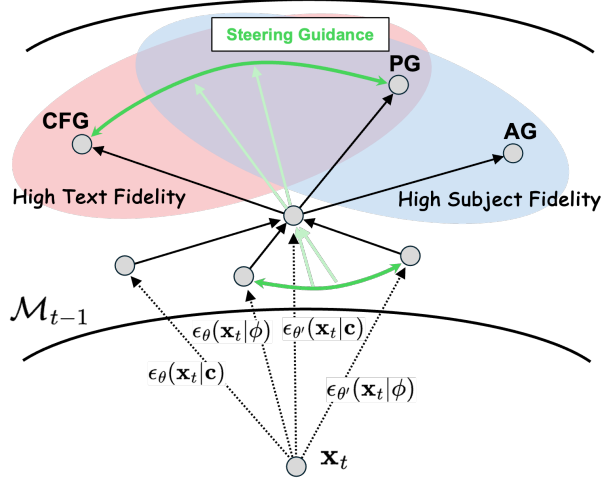


Figure 3. Comparison between Classifier-Free Guidance (CFG), AutoGuidance (AG), and our Personalization Guidance (PG), with the band conceptually representing the noisy data manifold. By leveraging weight interpolation, guidance can be steered toward the optimal space between CFG and PG.

$\epsilon_{\theta}(\mathbf{x}_t, \phi)$ and the conditional model $\epsilon_{\theta}(\mathbf{x}_t, \mathbf{c})$ within a single model by dropping $\mathbf{c}$ with certain probability to allow null conditioning with $\mathbf{c} = \phi$. In the personalization setting, parameterizing the score function with the fine-tuned diffusion model $\epsilon_{\theta'}$ is described as:

$$\tilde{\epsilon}_{\theta'}^{\lambda}(\mathbf{x}_t|\mathbf{c}) = \epsilon_{\theta'}(\mathbf{x}_t|\phi) + \lambda(\epsilon_{\theta'}(\mathbf{x}_t|\mathbf{c}) - \epsilon_{\theta'}(\mathbf{x}_t|\phi)). \quad (5)$$

However, while CFG improves alignment with the conditions, such as the text prompt, it struggles to enhance adaptation to the target data in the personalization task.

AutoGuidance. AutoGuidance (AG) [18] is another guidance technique that employs a weaker version of the same model to identify suboptimal generation trajectories. Rather than using the unconditional model $\epsilon_{\theta}(\mathbf{x}_t|\phi)$, AG directly guides a model with the weak model $\epsilon_{\theta_{weak}}$, which is trained on the same task and conditioning, but degraded by under-training or less-parameters. However, obtaining a weaker version $\epsilon_{\theta_{weak}}$ of a fully trained diffusion model like Stable Diffusion [26] is challenging in practice. Instead, we found that a less trained model could serve as the pre-trained model for personalization, where $\theta_{weak} = \theta$, effectively shifts the output away in the direction of fine-tuning, enhancing the adaptation to the target data. In the personalization, AG is formulated as:

$$\tilde{\epsilon}_{\theta'}^{\lambda}(\mathbf{x}_t|\mathbf{c}) = \epsilon_{\theta}(\mathbf{x}_t|\mathbf{c}) + \lambda(\epsilon_{\theta'}(\mathbf{x}_t|\mathbf{c}) - \epsilon_{\theta}(\mathbf{x}_t|\mathbf{c})). \quad (6)$$

However, this approach has a major drawback—it scarcely reflects the text prompt, as shown in Fig. 2.

3.2. Steering Personalization Guidance

Inspired by autoguidance [18], we simply modify the CFG [10], by leveraging a pre-trained diffusion model ϵ_{θ}

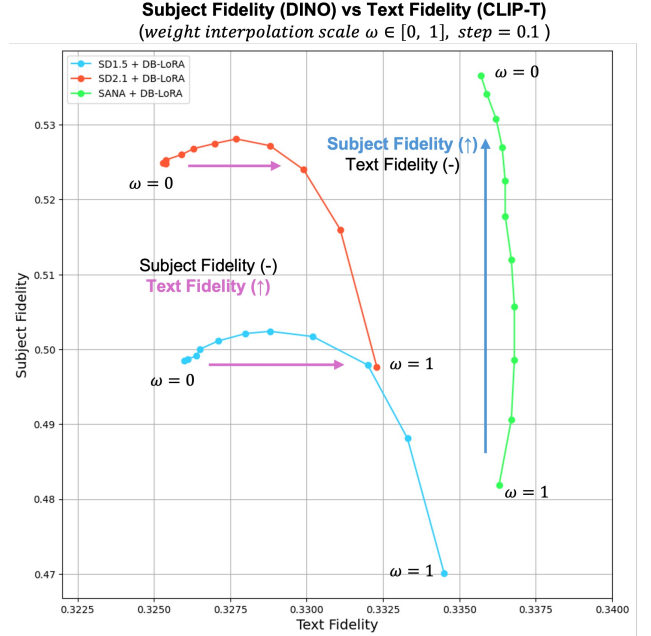


Figure 4. Ablation study on the weight interpolation scale $\omega \in [0.0, 1.0]$ with measurements taken at intervals of 0.1. Here, we use DreamBooth (DB) LoRA based on SD 1.5, SD 2.1, and SANA. By adjusting ω , it is possible to improve subject fidelity or text fidelity, while preserving the other.

instead of the fine-tuned model $\epsilon_{\theta'}$ as the weak model.

$$\tilde{\epsilon}_{\theta'}^{\lambda}(\mathbf{x}_t|\mathbf{c}) = \epsilon_{\theta}(\mathbf{x}_t|\phi) + \lambda(\epsilon_{\theta'}(\mathbf{x}_t|\mathbf{c}) - \epsilon_{\theta}(\mathbf{x}_t|\phi)). \quad (7)$$

Despite being a simple extension, we found that this approach significantly improves subject fidelity.

Fig. 3 shows the rationale behind our personalization guidance. Since this simply modified guidance can steer the output toward the intersection space where both text fidelity and subject fidelity are high. As illustrated in Fig. 3, we explore the feasibility of identifying a superior space within the noisy data manifold. We hypothesize that an optimal point for text and subject fidelity could be located between CFG and PG, and we propose a method to facilitate the steering of guidance. Steering the guidance involves the adjustment of an appropriate weak model. By identifying a suitable weak model, we can steer $\epsilon_{\theta}(\mathbf{x}_t|\phi)$ towards a more favorable point. We propose employing a weight interpolation to obtain an appropriate weak model. Specifically, by interpolating weights between the parameters of the pre-trained and the fine-tuned models, we can modulate the degree of adaptation to the target data, thereby obtaining an appropriate weak model $\epsilon_{\theta_{\omega}}$.

$$\theta_{\omega} = \omega \theta' + (1 - \omega) \theta, \quad \omega \in [0, 1], \quad (8)$$

where ω denotes the scale of weight interpolation. Increasing ω moves parameters closer to the fine-tuned model θ' ,

	Method	λ	DINO	CLIP-I	CLIP-T
SD 1.5	DB-LoRA	-	0.3741	0.6797	0.2834
	+ CFG	7.5	0.4701	0.7349	0.3345
	+ AG	2.0	0.4831	0.7408	0.3236
	+ Ours ($\omega = 0$)	7.5	<u>0.4985</u>	0.7516	0.3260
	+ Ours ($\omega = 0.6$)	7.5	0.5024	<u>0.7510</u>	<u>0.3288</u>
SD 2.1	DB-LoRA	-	0.4202	0.7076	0.2929
	+ CFG	7.5	0.4976	0.7519	0.3323
	+ AG	2.0	0.5072	0.7571	0.3212
	+ Ours ($\omega = 0$)	7.5	<u>0.5248</u>	<u>0.7655</u>	0.3254
	+ Ours ($\omega = 0.6$)	7.5	0.5281	0.7660	<u>0.3277</u>
SANA	DB-LoRA	-	0.4190	0.6939	0.3064
	+ CFG	4.5	<u>0.4819</u>	<u>0.7291</u>	0.3363
	+ AG	2.0	0.4792	0.7284	0.3355
	+ Ours ($\omega = 0$)	4.5	0.5366	0.7522	<u>0.3357</u>

Table 1. Comparison with other guidance methods across different base diffusion models including SD1.5, SD2.1, and SANA. Here, we use **DreamBooth-LoRA (DB-LoRA)** only.

reducing the loss $\mathcal{L}$ on $\mathcal{D}_{\text{target}}$. Finally, applying the weight interpolation to the weak model in the guidance:

$$\tilde{\epsilon}_{\theta'}^{\lambda}(\mathbf{x}_t|\mathbf{c}) = \epsilon_{\theta_{\omega}}(\mathbf{x}_t|\phi) + \lambda(\epsilon_{\theta'}(\mathbf{x}_t|\mathbf{c}) - \epsilon_{\theta_{\omega}}(\mathbf{x}_t|\phi)). \quad (9)$$

Note that $\omega = 1$ is equal to CFG. As shown in Fig. 4, it is possible to easily control the degree of subject and text fidelity. If θ includes only linear classifier, the personalization guidance with weight interpolation is equal to the combination of CFG and AG, which is the output-space interpolation. However, as neural networks are non-linear with respect to parameters, guidance with weight interpolation is different from the output interpolation of CFG and AG [36].

4. Experiments

4.1. Experimental Setting

Datasets. We conduct our experiments on the ViCo dataset [9], which includes 16 concepts and 31 text prompts. Each concept includes a small set of reference images (typically 4-7) and associated text prompts. Following the previous work [22], we fine-tune the pre-trained text-to-image diffusion models per concept and then generate 8 images per concept and text prompt, resulting in a total of 3,968 images for evaluation.

Evaluation Metrics. Following the existing personalization studies [14, 22, 27], we evaluate subject and text fidelity. For the quantitative assessment, we adopt three evaluation metrics: (i) CLIP-T score [25], which uses CLIP’s text-to-image similarity to evaluate text prompt alignment; (ii) CLIP-I score, which computes image-to-image similarity using CLIP embeddings to gauge how closely generated samples match reference images; and (iii) DINO score [3], which calculates image-to-image similarity between reference and generated images, offering a complementary perspective on subject fidelity.

	Method	λ	DINO	CLIP-I	CLIP-T
SD 1.5	DB-LoRA+TI	-	0.3725	0.6775	0.2817
	+ CFG	7.5	0.4618	0.7292	0.3316
	+ SAG	1.0	0.4600	0.7273	<u>0.3307</u>
	+ AG	2.0	0.4806	0.7408	0.3264
	+ Ours ($\omega = 0$)	7.5	<u>0.4814</u>	<u>0.7459</u>	0.3233
	+ Ours ($\omega = 0.7$)	7.5	0.4880	0.7461	0.3272
	ClassDiffusion	-	0.3703	0.6669	0.2757
	+ CFG	7.5	0.5287	0.7517	0.3305
	+ SAG	1.0	0.5172	0.7465	0.3296
	+ AG	2.0	0.5570	0.7628	0.3280
SD 2.1	+ Ours ($\omega = 0$)	7.5	0.5700	0.7656	<u>0.3297</u>
	DB-LoRA+TI	-	0.4514	0.7251	0.2823
	+ CFG	7.5	0.5291	0.7749	0.3244
	+ SAG	1.0	0.5328	0.7766	<u>0.3223</u>
	+ AG	2.0	0.5557	0.7877	0.3206
	+ Ours ($\omega = 0$)	7.5	<u>0.5683</u>	<u>0.7918</u>	0.3163
	+ Ours ($\omega = 0.6$)	7.5	0.5689	0.7919	0.3185
	ClassDiffusion	-	0.4273	0.6995	0.2843
	+ CFG	7.5	0.5840	0.7757	0.3277
	+ SAG	1.0	0.5783	0.7739	0.3244
	+ AG	2.0	0.6060	0.7856	<u>0.3246</u>
	+ Ours ($\omega = 0$)	7.5	0.6166	0.7896	<u>0.3246</u>

Table 2. Comparison with other guidance methods across different base diffusion models including SD1.5 and SD2.1. Here, we use **DreamBooth-LoRA (DB-LoRA) + Textual Inversion (TI)** and **ClassDiffusion**. Since these personalization methods fine-tune text embedding, subject-agnostic guidance (SAG) is also compared with our method.

Implementation Details. In our experiments, we utilize three pre-trained text-to-image diffusion models as the base models: Stable Diffusion 1.5 (SD 1.5) [26], Stable Diffusion 2.1 (SD 2.1) [26], and SANA-1.6B [38]. SD 1.5 and SD 2.1 generate 512×512 resolution images, which comprise U-Net based architectures. On the other hand, SANA is a recent text-to-image diffusion model, which is based on the diffusion transformer [23] that can generate images up to 4096×4096 resolution. In the experiments, SANA generates 1024×1024 images.

To fine-tune the diffusion models, we leverage several personalization techniques: (i) DreamBooth-LoRA (DB-LoRA) [13, 27], (ii) DB-LoRA + Textual Inversion (TI), and (iii) ClassDiffusion [14]. We fine-tune each model for up to 500 steps on the reference images from the ViCo dataset. We set the number of sampling steps (*e.g.*, DDIM) between 20-50. Further details regarding additional hyperparameters are described in the supplementary.

Baselines. We evaluate our guidance method against other guidance techniques: (i) Classifier-Free Guidance (CFG) [10], (ii) Subject-Agnostic Guidance (SAG) [4], and (iii) AutoGuidance (AG) [18]. Since SAG and AG alone struggle to generate images that accurately reflect the text prompt, they are used in conjunction with CFG to produce

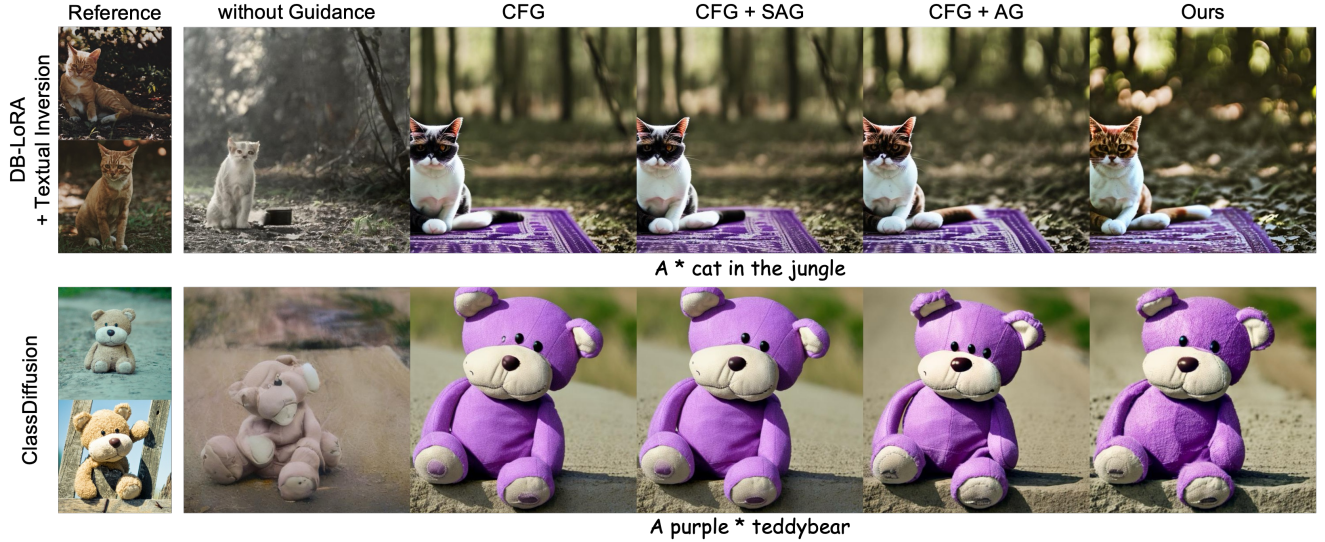


Figure 5. Comparison with other guidance techniques, including without guidance, CFG, CFG+SAG, and CFG+AG. These images are generated by the fine-tuned SD 2.1 using DB-LoRA and ClassDiffusion.



Figure 6. Comparison with other guidance techniques. These images are generated by fine-tuned SANA [38] using DB-LoRA.

images. Specifically, AG experiments are conducted by referring to recent work [19] that combines CFG and AG.

4.2. Results

Quantitative Results. Table 1 and Table 2 showcase the personalization performance on the ViCo dataset. Table 1 provides the quantitative results for DB-LoRA based on SD 1.5, SD 2.1, and SANA. Since DB-LoRA+TI and ClassDiffusion involve fine-tuning textual embeddings, we also evaluate subject-agnostic guidance (SAG) [4], which necessitates learned textual embeddings for specific concepts. As illustrated in both Table 1 and Table 2, our guidance method significantly enhances subject fidelity (DINO and CLIP-I scores), while maintaining text fidelity (CLIP-T score) with minimal change. Additionally, by adjusting ω , the model can further optimize personalization performance across all

evaluation metrics. We select the optimal ω based on the best DINO and CLIP-I scores. For SANA with DB-LoRA and ClassDiffusion, the best subject fidelity was observed at $\omega = 0$. These results demonstrate that our approach significantly improves subject fidelity while maintaining text fidelity, and that steering the guidance enables a balanced trade-off between the two.

Interestingly, the impact of personalized guidance varies depending on the personalization method. ClassDiffusion [14] employs a semantic preservation loss that maintains the text embedding space, leading to better overall text fidelity retention. As a result, our guidance method maintains the CLIP-T score more effectively when applied to ClassDiffusion than to other personalization methods. This observation highlights a key insight: the better the fine-tuning direction, the more effective our guidance becomes.

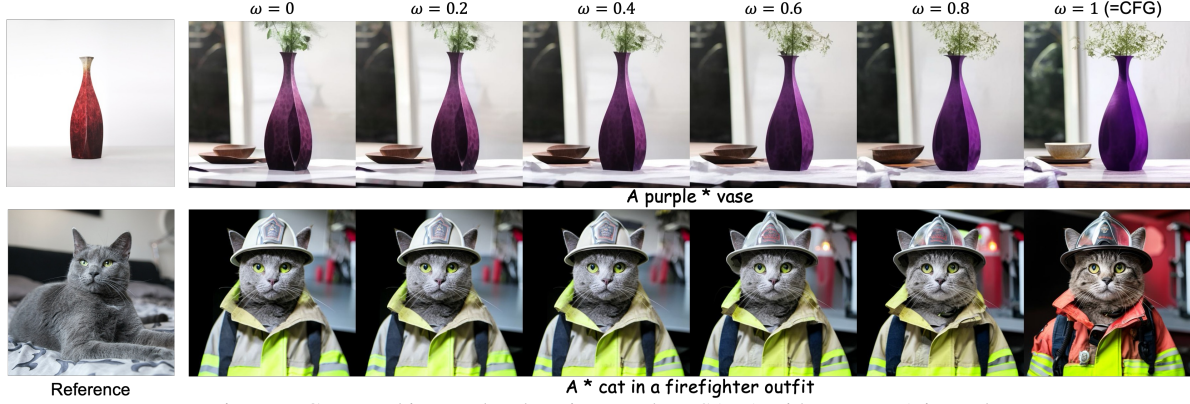


Figure 7. Generated images by changing ω , where SANA with DB-LoRA is used.

Method	Subject Fidelity $\uparrow$	Text Fidelity $\uparrow$
Stable Diffusion 2.1 + ClassDiffusion		
CFG	10.45%	15.24%
CFG + SAG	8.01%	6.20%
CFG + AG	12.60%	6.20%
Ours	55.22%	32.04%
Undecided	13.67%	40.31%
SANA + DB-LoRA		
CFG	2.48%	3.08%
CFG + AG	2.17%	2.77%
Ours	68.32%	8.64%
Undecided	27.01%	85.49%

Table 3. User preference on subject fidelity and text fidelity.

Qualitative Results. Fig. 5 and Fig. 6 show the comparison results based on SD 2.1 and SANA, respectively. As shown in Fig. 5, CFG often fails to capture fine details of the reference subject (e.g., cat’s appearance and the teddybear’s eye). Additionally, applying SAG or AG has minimal impact in some cases (first row of Fig. 5) or even degrades image quality, introducing artifacts such as a teddy bear with three eyes (CFG+AG, second row of Fig. 5). In contrast, our method consistently achieves better subject fidelity with the same noise input. Similarly, in Fig. 6, our method better preserves subject details compared to CFG and CFG+AG, as evidenced by the significant improvement in subject fidelity in quantitative results. For instance, it accurately retains the ears and color of a dog (first row of Fig. 6) and the shape and numerical details of a clock (second row of Fig. 6). In contrast, applying AG with CFG shows little to no improvement despite incurring a higher computational cost compared to both CFG and our guidance method.

User Study. Table 3 presents the user preference results comparing our method with baseline approaches. The results show that our guidance technique achieves significantly higher subject fidelity than other guidance methods. Our approach also shows superior text fidelity in user evaluations. Further details can be found in the supplementary.

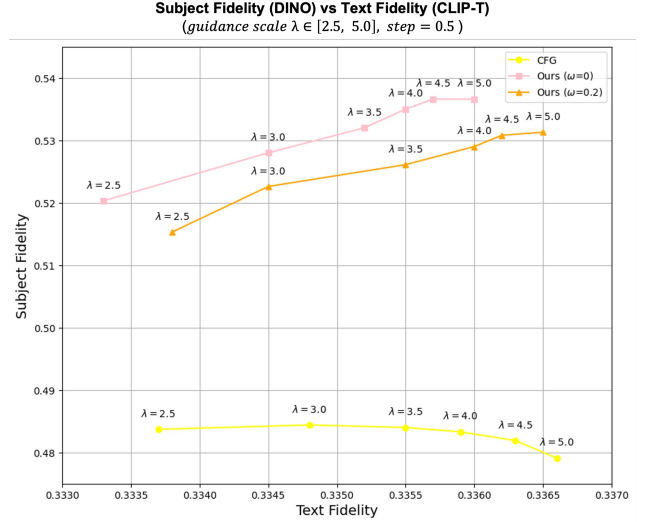


Figure 8. Ablation study on the guidance scale $\lambda \in [2.5, 5.0]$, measuring performance at 0.5 intervals using DB-LoRA based on SANA. When ω is set to 0.2, we can improve both subject and text fidelity compared to CFG, regardless of guidance scale λ . Note that CLIP-T score is slightly changed in the graph ($\max(\text{CLIP-T}) - \min(\text{CLIP-T}) < 0.004$).

4.3. Further Analysis

Ablation Study on ω . Fig. 4 illustrates how DINO and CLIP-T scores vary with changes in ω . For both SD 1.5 and SD 2.1, personalization guidance slightly reduces text fidelity but significantly enhances subject fidelity compared to CFG ($\omega = 1$). By appropriately adjusting ω , it is possible to either boost text fidelity while maintaining subject fidelity (as indicated by the pink arrow) or enhance subject fidelity while preserving text fidelity (as indicated by the blue arrow). It is important to note that these trends may vary depending on the personalization technique or the pre-trained diffusion model. However, the ability to easily fine-tune results via weight interpolation at inference, without modifying training steps or learning rates, offers a major practical advantage over traditional methods for controlling overfitting. Fig. 7 further visualizes the impact of ω on gen-

Method	PickScore $\uparrow$	Method	DreamSim $\downarrow$
DPO+CFG	21.53	PairCustom+CFG	0.7993
DPO+Ours	21.58	PairCustom+Ours	0.7344

Table 4. Results of combining Diffusion-DPO with ours. Table 5. Results of combining PairCustom with ours.



Figure 9. Qualitative results of our guidance across diverse tasks.

eration quality. The first row shows that higher ω values retain fine details, such as the intricate patterns on the vase, at the cost of weaker adherence to the "purple" text prompt. Additionally, as ω decreases, subject fidelity improves significantly, particularly in the range of 0.6 to 1.0.

Ablation Study on Guidance Scale. Fig. 8 shows how personalization performance varies with different guidance scales λ . We evaluate our guidance at $\omega = 0$ and $\omega = 0.2$, where text fidelity remains preserved. As λ decreases, text fidelity deteriorates for both guidance methods. However, as λ increases, our guidance method shows improvements in both the DINO score and CLIP-T score, whereas the DINO score of CFG declines. This suggests that CFG reduces subject fidelity, and adjusting the guidance scale alone is insufficient to correct this effect.

4.4. Application to Fine-tuned Diffusion Models.

Fine-tuning diffusion model extends beyond personalization tasks to applications such as style personalization [17], human preference learning [21, 33], and specific task learning [2, 15]. Our personalization guidance framework leverages the knowledge gap between the pre-trained and fine-tuned models, emphasizing the information better captured by the fine-tuned model. This concept can be generalized to control the degree of adaptation for other fine-tuned models, such as adjusting the extent of human preference alignment

or style personalization during inference.

To this end, to investigate our method’s generalizability across diverse tasks, Fig. 9 show the following experiments: (a) Human Preference Optimization (Diffusion-DPO [33]), (b) Style Personalization (PairCustom [17]), and (c) Instruction-based Editing (InstructPix2Pix [2]).

(a) Human Preference Optimization: For evaluation with the Diffusion-DPO model [33], we use the PartiPrompts dataset [39] and report PickScore [39], which reflects human-perceived image quality. Table 4 shows that integrating our method into the SD 1.5-based Diffusion-DPO model yields superior performance compared to CFG, confirming the effectiveness of our approach on human preference optimization tasks. As shown in Fig. 9 (a), our method helps generate more natural-looking images and corrects artifacts that would otherwise appear unnatural (*e.g.*, additional finger).

(b) Style Personalization: To evaluate style alignment, we use two pre-trained style-LoRAs released by PairCustom, combining each with both CFG and our method. Using text prompts from the StyleAlign [37] paper, we generate 800 images and compute the DreamSim score [7] to quantify the similarity between the target style and generated outputs. As shown in Table 5, applying our method significantly improves style alignment, demonstrating its effectiveness for style personalization. In Fig. 9 (b), combining PairCustom with our guidance leads to a more faithful transfer of the painting style from the reference image.

(c) Instruction-based Editing: In Fig. 9 (c), our guidance improve the performance of the InstructPix2Pix [2], which is designed to edit images based on text instructions, to better follow the given instruction. As a result, the generated outputs are more accurately aligned with the intended edits.

These results demonstrate that applying our guidance to various fine-tuned diffusion models allows them to generate images more faithfully aligned with their respective fine-tuning objectives. This highlights the generalizability of our method across diverse tasks. Further implementation details are provided in the supplementary.

5. Conclusion

This paper tackles the critical challenge of balancing adaptation to the target distribution and knowledge preservation from the original model in personalized diffusion models. Our approach simply modifies the existing guidance by adjusting the degree of unlearning in the weak model through weight interpolation between the pre-trained and fine-tuned models. This simple yet effective guidance enhances subject fidelity while preserving text editability, ensuring an optimal trajectory for personalization. Furthermore, we demonstrate that our guidance can be effectively applied to a wide range of fine-tuned diffusion models, such as human preference learning.

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